

Probabilistic Sufficiency for the Obstruction Calculus: Belief-State Safety Values under Partial Observation

Amin Abaee

Independent Researcher

ORCID: 0000-0002-0019-1842

amin_abaee@ut.ac.ir

September 26, 2026

Abstract

The obstruction calculus characterizes nonviability through certificates on information states; the probabilistic side asks what the certificates imply when the information state is a distribution. This paper develops the belief-state safety value $V_k(b)$ — the maximal probability of remaining safe for k steps — as a theory carried entirely in exact rational arithmetic, with six results. First, a support identity: on finite models with deterministic kernels and deterministic observation maps the posterior support coincides with the set-valued post-state of the calculus, the value-one level of the recursion coincides with the viable-set recursion $\mathcal{W}_k$, and outside it the deficit satisfies $1 - V_k(b) \geq \min_{x \in \text{supp}(b)} b(x)$, attained on natural instances; moreover the alpha-vectors are exactly the indicators of the maximal jointly survivable subsets of the support — an antichain, Sperner-bounded — which explains why the audited witness census realizes three types. Second, exactness and parametric closed forms: over rational input data the finite-horizon values are piecewise linear with rational alpha-vectors, and on the delayed hidden-regime class both the declared hold-class and the open-loop values are proved in closed form for all initial states, all window lengths and horizons, and all prior weights — value one exactly on $z_0 \geq 1 + \min(k, T_{\text{obs}})$ and on $z_0 \geq 2$ respectively, the larger prior mass off those regions, zero below the floor — so the complete 16×12 audited value table, the 36/156 census, and the 384 hold-direction witnesses (348 distinct after deduplication) become one-line corollaries. Third, the class lattice: declared hold $\subseteq$ open-loop $\subseteq$ sequential $\subseteq$ fully observed orders the values, and both gaps are realized exactly — the value of flexibility on the timing cells and the value of observation on the common-action cells, each $\frac{1}{2}$ at the symmetric prior and zero elsewhere. Fourth, agreement and stabilization: the segment-form enumeration reproduces the closed forms on all 48 audited cells and horizons $k \leq 4$, the certificate partition coincides with the value boundaries, the hold-class value jumps by exactly $\frac{1}{2}$ at $z_0 = 1 + \min(k, T_{\text{obs}})$, the two classes differ exactly where $2 \leq z_0 < 1 + \min(k, T_{\text{obs}})$ (on the audited grid, exactly the timing cells, 36 combinations), and the value is frozen once $k \geq T_{\text{obs}}$ on the finite survivable-set lattice, so the audited tables are complete. Fifth, a genuinely stochastic layer: an exact weighted-path deficit identity under noisy observation — the deficit is

the prior mass weighted by fatal-reading probabilities, and a two-state sensor instance attains deficit $1/20$ against a minimal mass of $1/2$, so the min-mass bound is not universal there — a noisy-probe deadline family with three-valued exact values, and the repetition trade-off: a second probe cuts the misread probability from $1/10$ to $7/250$, and each probe step costs $1/10$ of floor margin. Sixth, a distributionally robust interpolation between the expected and adversarial operators over contamination ambiguity sets — V^ρ is the exact ρ -mixture of the two, recovering expectation at $\rho = 0$ and the worst-case reading at $\rho = 1$, with an exact closed form on the delayed class — an exact probe-count bound $p_{\text{wrong}}(n) \leq (4\varepsilon(1 - \varepsilon))^{n/2}$ giving the $\log(1/\delta)/\text{sep}^2$ probe law, and the general additive deadline law: the threshold is the floor plus the probe excursion plus drift times deadline, verified on two instances. Every closed form is verified against brute-force enumeration from the definitions, including a seeded randomized battery.

Keywords: partially observable control; viability kernels; belief states; safety values; exact arithmetic; adaptive observation

1. Introduction

Viability under incomplete observation divides into a necessity side and a sufficiency side. The obstruction calculus treats the necessity side: certificates whose firing proves nonviability of an information state (Abaee, 2026b). The probabilistic question is the quantitative counterpart. If the controller holds a distribution b over states rather than a set, how safe is safe? The answer is a function: the belief-state safety value $V_k(b)$, the maximal worst-case probability of keeping the state trajectory in the safe set for k steps. Its value-one level is the viability statement of the set-valued theory; everything below level one is measured by the deficit $1 - V_k(b)$, and the obstruction certificates reappear as lower bounds on that deficit.

This paper develops the probabilistic layer of the viability theory of Abaee (2026b): the calculus there is set-valued, and the present theory quantifies what remains safe once the information state is a distribution. It brings together the stochastic selector development, the belief-state value campaign, and the hidden-parameter learning-deadline audit into one theory with parametric closed forms, a class lattice, and an exact stochastic layer; the

companion papers (the calculus, its computational certification, and the worked-systems audit) are unaffected, and nothing verified in them is re-proved differently here.

The development proceeds in seven steps. Section 2. states the augmented model — total transition rows, an absorbing unsafe state carrying its own observation, masked witnesses — and the value recursion, distinguishing the selector’s attaining set from its value-one level. Section 3. connects the belief-state recursion to the set-valued recursion of the calculus exactly, through the support of the belief. Section 4. shows the finite-horizon values are piecewise linear and exactly representable, and carries the complete audited value table over the delayed grid. Section 5. evaluates the deterministic limit, at which the recursion degenerates to a survived-mass formula over blind action sequences, with the asymmetric audit carried sequence by sequence. Section 6. reads the obstruction certificates systematically as deficit lower bounds. Section 7. proves agreement between the declared-class segment enumeration and the closed-form values on the audited delayed-regime class, quantifies what the class declaration contributes, locates the value’s jump loci, and carries the derivations. Section 8. treats observation as a design variable: probes, learning deadlines, and the exact kernel cost of late resolution of a hidden parameter.

Relation to prior work. The belief-support reduction at the core of Theorem 3 is the finite-horizon qualitative analysis of partially observable Markov decision processes: almost-sure safety depends only on belief supports, as established for qualitative objectives by Chatterjee, Doyen, and Henzinger (2009). What is added here is quantitative and structural: the deficit bound and its witness reading (Propositions 7 and 12), the survivable-set (antichain) formula behind the witness census (Proposition 4), the class lattice with its realized gaps (Theorem 14), the parametric closed forms (Proposition 5 and Theorem 6), and the class-scoped timing obstruction. On complexity, finite-horizon partially observable problems are PSPACE-complete and their no-observation variant NP-complete (Papadimitriou and Tsitsiklis, 1987); the exponential witness-growth law of Proposition 7 is the honest general behaviour, and the audited two-witness families are structural rather than representative. Runtime safety filters for learning agents (Alshiekh et al., 2018) and distributionally robust partially observable models over ambiguity sets (Nakao, Jiang, and Shen, 2021), whose value functions remain piecewise linear, bound the present development from the computational and the modelling side; Section 12. records the ambiguity-set interpolation as the formal home of worst-case wording.

All audited computations are exact: integer and rational arithmetic, standard library only, deterministic scripts (Section 11.).

2. The augmented model, the safety value, and the selector

The augmented state space. Let the physical safe set be $\mathcal{V}$ and let the state space of the model be $X^\perp = \mathcal{V} \cup \{\perp\}$, where $\perp$ is an absorbing unsafe state. For $x \in \mathcal{V}$ the transition row splits between $\mathcal{V}$ and $\perp$,

$$\sum_{x' \in \mathcal{V}} T(x' | x, a) \leq 1, \quad T(\perp | x, a) = 1 - \sum_{x' \in \mathcal{V}} T(x' | x, a),$$

and the bottom row is $T(\perp | \perp, a) = 1$: every row of T sums to one and $\perp$ is absorbing, so exit from $\mathcal{V}$ is absorption and no trajectory re-enters $\mathcal{V}$ from outside it. The observation model is completed on all of $X^\perp$: there is a distinguished observation $y_\perp$ with $g(y_\perp | \perp, a, \perp) = 1$ (and $g(y | \cdot, \cdot, \perp) = 0$ for $y \neq y_\perp$), while $g(\cdot | x, a, x')$ on $\mathcal{V}$ is arbitrary non-negative with the usual normalization per (x, a) . A belief is a distribution b over $X^\perp$, updated by Bayes’ rule $\tau(b, a, y)$, with $\mathbb{P}(y | b, a) = \sum_{x, x'} b(x) T(x' | x, a) g(y | x, a, x')$; because every row sums to one and every (x, a, x') emits exactly one observation with probability one, the laws $\mathbb{P}(y | b, a)$ sum to one over y at every belief — no sub-probability bookkeeping is needed. On the deterministic kernels of the audited instances this formalism converts set-valued nonviability into almost-sure nonviability on the augmented chain.

Definition 1 (safety value). Fix a declared policy class Π . The safety value at horizon k is the maximal class-survival probability

$$V_k^\Pi(b) = \max_{\pi \in \Pi} \mathbb{P}_\pi(x_t \in \mathcal{V} \text{ for } t = 0, \dots, k | b),$$

the horizon counting the current state. The unrestricted class $\Pi = \Pi_{\text{seq}}$ is the class of all sequential policies.

The selector and its levels. In the selector framework this is the distributional counterpart of the recursive correspondence: the information state is b , a response is a class-admissible policy segment. Two objects must be distinguished. The *attaining set* at b ,

$$\Gamma_k^\Pi(b) = \left\{ a \in A : \sum_y \mathbb{P}(y | b, a) V_{k-1}^\Pi(\tau(b, a, y)) = V_k^\Pi(b) \right\},$$

collects the actions attaining the value; for finite A it is nonempty at every belief and horizon — nonemptiness is automatic and carries no viability content. The *value-one level*,

$$S_k^\Pi(b) = \left\{ a \in A : \sum_y \mathbb{P}(y | b, a) V_{k-1}^\Pi(\tau(b, a, y)) = 1 \right\},$$

is the selector’s viable-response set, and its nonemptiness corresponds exactly to $V_k^\Pi(b) = 1$. Obstruction certificates correspond to deficits $1 - V_k^\Pi(b) > 0$. Notation is fixed as follows: Γ_k^Π denotes alpha-vector witness sets, $\Gamma_k^\Pi(b)$ the attaining action set, $\mathcal{W}_k$ the calculus’s viable-set recursion (a family of state sets), and $\mathcal{W}_k^{\text{bel}}$ its belief-space analogue (a family of beliefs); unsuperscripted values do not appear.

Theorem 2 (belief-state value recursion). $V_0^\Pi(b) = b(\mathcal{V})$ (the horizon counts the current state), and for $k \geq 0$,

$$V_{k+1}^\Pi(b) = \max_{a \in A(\Pi)} \sum_{y \in Y} \mathbb{P}(y | b, a) V_k^\Pi(\tau(b, a, y)),$$

where $A(\Pi)$ is the class-admissible action set at the current stage for Markov classes Π . The recursion is exact on every finite horizon; over the unrestricted class it is the Smallwood–Sondik belief-state value iteration specialized to the absorbing safety reward (Smallwood and Sondik, 1973).

The proof is standard value iteration for the belief-state Markov decision process, with the absorbing state making V_k^Π the k -step class-survival probability (Bertsekas and Shreve, 1978). For non-Markov classes (the hold and open-loop classes of Section 7.) the class constraint couples stages; the audited computations there use the equivalent segment form of Proposition 7 below.

3. The support identity and exact sufficiency

The bridge to the set-valued theory runs through the support. Write $\text{supp}(b)$ for the states carrying positive mass,

and let $\mathcal{W}_k$ denote the calculus's viable-set recursion: $\mathcal{W}_0 = \{B \subseteq \mathcal{V}\}$ and $\mathcal{W}_k = \{B : \text{some admissible } a \text{ maps every } x \in B \text{ into } \mathcal{W}_{k-1}\text{-supported posteriors}\}$.

Theorem 3 (support identity). *On finite models with deterministic kernels and deterministic observation maps, for every action a and observation y ,*

$$\text{supp}(b^+(\cdot | a, y)) = \text{Post}(\text{supp}(b), a, y),$$

the set-valued post-state of the calculus. Consequently, for the unrestricted sequential class, $V_k(b) = 1$ if and only if $\text{supp}(b) \in \mathcal{W}_k$, while if $\text{supp}(b) \notin \mathcal{W}_k$ then every policy loses a compatible state of positive mass along its realized observation path (the loss can occur at any stage), so

$$1 - V_k(b) \geq \min_{x \in \text{supp}(b)} b(x).$$

Moreover $V_k(b)$ is non-increasing in k .

Proposition 4 (survivable sets, witnesses, and the census). *Let the kernels and observation maps be deterministic, fix a declared class, and let $\mathcal{S}_k$ be the family of subsets S of the support that are jointly survivable by one admissible class-element (sequence or policy). Then: (i) $V_k(b) = \max_{S \in \mathcal{S}_k} b(S)$; (ii) the alpha-vectors are exactly the indicators of the maximal elements of $\mathcal{S}_k$ in the componentwise order; (iii) the maximal elements form an antichain, so their number is at most $\binom{n}{\lfloor n/2 \rfloor}$ for a support of size n (Sperner); (iv) the deficit is $1 - V_k(b) = \min b(S^c)$, the minimum over the maximal survivable sets, and the min-mass bound is the special case $|S^c| = 1$. On the delayed instance the maximal survivable sets are $\{+, -\}$ exactly when $z_0 \geq 2$ and the two singletons otherwise (at $z_0 \geq 1$), giving the empty, mirror-pair, and merged witness families — the three-type census of the audited campaign is this antichain read off the grid.*

Branches survive a fixed sequence independently, survived prior mass sums over surviving branches, and maximizing over class-elements gives (i); the indicator of a survivable set is a witness and dominance gives (ii) with deduplication; maximality gives (iii) with Sperner's theorem; and monotonicity of $b(S)$ in S gives (iv).

Proposition 5 (closed form on the delayed class, declared hold class). *On the delayed hidden-regime instance — states (z, θ) with $\theta \in \{-1, +1\}$, prior $b_0 = (\frac{1}{2}, \frac{1}{2})$ from z_0 , no observation before T_{obs} , exact revelation at T_{obs} , and unit drift against the floors — the belief support follows the two branches $z_0 \mp k$, and the declared hold class value (Π_{hold} : one blind hold $u \equiv \pm 1$ for the whole window) is*

$$V_k^{\Pi_{\text{hold}}}(b_0) = \begin{cases} \frac{1}{2} (\mathbb{K}[z_0 - k \geq 1] + \mathbb{K}[z_0 + k \geq 1]), & k \leq T_{\text{obs}}, \\ \frac{1}{2} (1 + \mathbb{K}[z_0 \geq 1 + T_{\text{obs}}]), & k > T_{\text{obs}}. \end{cases}$$

In particular $V_k^{\Pi_{\text{hold}}}(b_0) = 1$ exactly on $z_0 \geq 1 + T_{\text{obs}}$, recovering the declared-class deterministic verdict as the value-one level; Theorem 15 quantifies where the unrestricted class does better. This is a statement about the declared class, proved by direct enumeration below — not a corollary of Theorem 3, whose value is the unrestricted one.

Theorem 6 (parametric closed forms). *On the delayed hidden-regime instance with prior mass w on the $\theta = +1$ branch (arbitrary rational $w \in [0, 1]$), initial state z_0 (arbitrary real), and effective window $\ell = \min(k, T_{\text{obs}}) \geq 1$:*

$$V_k^{\Pi_{\text{hold}}}(b_0) = \begin{cases} 0, & z_0 < 1, \\ 1, & z_0 \geq 1 + \ell, \\ \max(w, 1 - w), & 1 \leq z_0 < 1 + \ell, \end{cases}$$

$$V_k^{\Pi_{\text{ol}}}(b_0) = \begin{cases} 0, & z_0 < 1, \\ 1, & z_0 \geq 2, \\ \max(w, 1 - w), & 1 \leq z_0 < 2. \end{cases}$$

For $k = 0$, $V_0(b_0) = \mathbb{K}[z_0 \geq 1]$. At the symmetric prior both off-viability values are $\frac{1}{2}$, and Proposition 5, the complete audited table, the 36/156 census, and the witness types are one-line corollaries (the grid has $z_0 \geq 1$, so the zero level is vacuous there).

Proof. Survival requires the state in the safe set at every stage, in particular at $t = 0$: if $z_0 < 1$ both branches are below the floor immediately and the value is zero. Under a hold u both branches move monotonically, so each survives the window exactly if it is at or above the floor at its horizon position: under $u = +1$ the $+$ branch rises (safe) and the $-$ branch ends at $z_0 - \ell$, giving survived mass $w + (1 - w)\mathbb{K}[z_0 \geq 1 + \ell]$; the mirror holds for $u = -1$; maximizing over the two holds gives the first display. Under open-loop play the partial sums of any ± 1 -sequence move the branches in opposition, so the first move sends one branch to $z_0 - 1$ and joint survival requires $z_0 \geq 2$; the alternating sequence keeps both branches in $[z_0 - 1, z_0 + 1]$, which lies above the floor exactly when $z_0 \geq 2$; a single branch is survivable by the constant sequence away from it (which makes that branch rise), so off the joint region the value is the larger prior mass. $\square$

Derivation of Proposition 5. While $k \leq T_{\text{obs}}$ no observation arrives, so a declared-class policy commits to one blind hold $u \equiv \pm 1$: under $u \equiv +1$ the $\theta = +1$ branch sits at $z_0 + k$ at the horizon and the $\theta = -1$ branch at $z_0 - k$; under $u \equiv -1$ the roles swap. Each branch carries mass $\frac{1}{2}$ and survives exactly when it stays at or above the floor, giving $\frac{1}{2}(\mathbb{K}[z_0 - k \geq 1] + \mathbb{K}[z_0 + k \geq 1])$, and the expectation over the prior retains exactly the surviving mass — the lost branch simply carries the missing half; no adversarial selection is involved. For $k > T_{\text{obs}}$, revelation at T_{obs} splits the belief: the posterior collapses to the realized branch, and the rising control $u = +1$ keeps it safe exactly when it sits at or above the floor at revelation — so survival reduces to surviving the blind window of length T_{obs} on both branches, the mass- $\frac{1}{2}$ falling branch sitting at $z_0 - T_{\text{obs}}$ at revelation, whence the indicator $\mathbb{K}[z_0 \geq 1 + T_{\text{obs}}]$ weighting the certainly-saved mass $\frac{1}{2}$. (The rising continuation is the point: a falling continuation would exit later, and Proposition 11 pins the frozen value for all horizons.) $\square$

4. Exact piecewise-linear values

Proposition 7 (rational alpha-vectors, class-restricted). *Fix a declared class Π . At every horizon k , V_k^{Π} is piecewise linear and convex: $V_k^{\Pi}(b) = \max_{\alpha \in \Gamma_k^{\Pi}} \alpha^{\top} b$ for a finite witness set Γ_k^{Π} on $\mathbb{Q}^{|\mathcal{V}|+1}$ (coordinates indexed by $\mathcal{V} \cup \{\perp\}$), constructed by the masked backup*

$$\alpha^{a, \gamma} (x) = \sum_{x'} T(x' | x, a) \sum_y g(y | x, a, x') \gamma_y(x'), \quad \gamma_y \in \Gamma_k^{\Pi},$$

with $\Gamma_0^{\Pi} = \{\mathbf{1}_{\mathcal{V}}\}$ and the boundary condition $\alpha(\perp) = 0$ applied at every backup (the backup acts on $\mathcal{V}$ only; the $\perp$ -coordinate of every witness is anchored at zero and remains zero under the backup) and $\Gamma_{k+1}^{\Pi} = \{\alpha^{a, \gamma} : a \in A(\Pi), \gamma, a \text{ selection}\}$, duplicates removed by exact equality. For a blind open-loop window the equivalent segment form applies: each admissible action segment $(u_0, \dots, u_m) \in \Pi$ contributes the witness $\alpha^{\text{seg}} = (\mathbf{1}[\text{the segment saves branch } x])_x$ and $V_k^{\Pi}(b) = \max_{\text{seg}} \alpha^{\text{seg} \top} b$. Over rational T , g , and priors, every α is an exact rational vector, every $V_k^{\Pi}(b)$ at a rational belief is an exact rational, and the attaining witness is part of the computation. The worst-case growth is $|\Gamma_{k+1}^{\Pi}| \leq |A| |\Gamma_k^{\Pi}|^{|\mathcal{V}|}$; deduplication and pruned

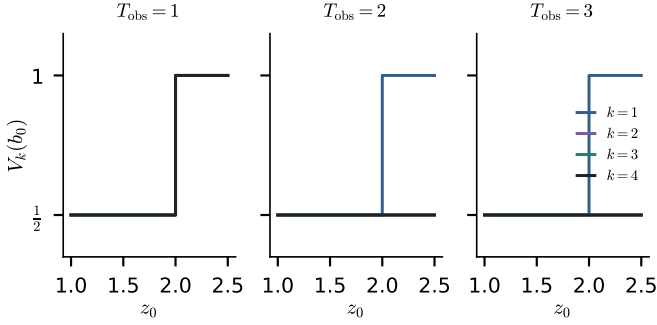


Figure 1: The closed-form values of Proposition 5 on the delayed grid: flat at $\frac{1}{2}$ while a branch falls, jumping by exactly $\frac{1}{2}$ at $z_0 = 1 + \min(k, T_{\text{obs}})$, constant beyond.

selections are the practical controls, and the exponential law is the honest complexity statement for the general class.

The PL structure is Smallwood–Sondik’s; the masking factor anchors $\alpha(\perp) = 0$ (survival requires the current state in $\mathcal{V}$, and $\perp$ is absorbing, so no backup can revive a lost branch). The exact-arithmetic discipline is the point: the backup maps rational vectors to rational vectors by finite sums and products, selections are finite, and equality tests for deduplication are exact on rationals. In the verification campaign the delayed grid enumerates the two hold-direction witnesses at each of its 192 cell-horizon combinations — 384 exact rational vectors (the two-floor instance’s witness set is carried separately) — and 348 distinct vectors remain after exact-equality deduplication.

Remark 8 (dimension). On the two-floor instance the value function on the one-dimensional simplex is $\max(b_0, b_1)$ for all audited horizons (Figure 5); the witness alpha-vectors live in the two-coordinate display, with the suppressed bottom coordinate identically zero — padded to $(1, 0, 0)$, $(0, 1, 0)$ on $\mathcal{V} \cup \{\perp\}$.

On the delayed grid the campaign evaluates $\alpha_k(z_0, T_{\text{obs}})$ at all $16 \times 3 \times 4 = 192$ cell-horizon combinations of $z_0 \in \{1.0, 1.1, \dots, 2.5\}$, $T_{\text{obs}} \in \{1, 2, 3\}$, $k \in \{1, 2, 3, 4\}$. The realized vectors take exactly three types: $(1, 1)$ (both branches savable in the blind window), 72 occurrences; and the mirror pair $(1, 0)$ and $(0, 1)$ (one branch savable), 156 occurrences each — the symmetry between the two mirror types is exact, $72 + 156 + 156 = 384$. The type counts are the pre-deduplication census over the two hold directions: on the 36 combinations where both holds save both branches the two witnesses coincide at $(1, 1)$ ($2 \times 36 = 72$), and exact-equality deduplication merges them, leaving $36 + 156 + 156 = 348$ distinct vectors — one per unit-value combination, two per the others. Table 1 carries the complete value table; Figure 1 displays the closed forms.

5. The deterministic limit and the two operators

Remark 9 (the two operators). The expectation over the disturbance law and the adversarial reading of its support are different Bellman operators. Proposition 10 states the deterministic degeneration as a change of operator: when the kernels are deterministic the stochastic expectation is replaced by the robust (adversarial-support) reading, and the resulting value is the robust value of the companion papers. The degeneration is a theorem about the robust operator; it is not a claim about the stochastic plant, whose expectation and robust readings differ in general (Section 12.).

Proposition 10 (survived-mass formula). *Let the kernels be deterministic, $x^+ = F(x, a, d)$, and let the disturbance be read*

adversarially within its support — the robust operator of Remark 9. Let the window be blind with declared sequential class Π_B . Then the degenerated value is the survived-mass formula

$$V_k(b) = \max_{(u_0, \dots) \in \Pi_B} \sum_{x \in \text{supp } b} b(x) \mathbf{1}[x \text{ survives } (u_0, \dots)],$$

a maximum of rational linear functions of b (the surviving indicators are the degenerate alpha-vectors). Consequently: if the support B carries a jointly surviving blind sequence, $V_k(b) = 1$; and if every declared sequence loses some branch, then $V_k(b) \leq 1 - \min_{x \in B} b(x)$ — the robust common-action verdicts and the minimal-mass deficit bound of the companion papers, recovered as exact values of the robust operator.

With deterministic kernels the posterior support follows the set-valued recursion, survival from a branch under a fixed sequence is a deterministic event, and prior mass sums over surviving branches; maximizing over declared sequences gives the formula, from which both consequences are immediate (a lost branch carries at least the minimal mass).

Proposition 11 (stabilization on the survivable-set lattice). *With deterministic kernels the survivable-set families satisfy $\mathcal{S}_{\ell+1} \subseteq \mathcal{S}_\ell$; hence $V_\ell(b)$ is non-increasing in the horizon and stabilizes after at most $2^{|\text{supp } b|}$ strict decreases, the frozen value being $\max_S b(S)$ over the maximal sets of the frozen family — an exact computable infinite-horizon value. On the delayed instance the open-loop family is frozen from $\ell = 1$ (oscillation and single-branch rises are length-independent), and the declared class freezes once $k \geq T_{\text{obs}}$ (the window ends and the rising continuation $u = +1$ then preserves safety indefinitely above the floor); consequently every audited table at $T_{\text{obs}} \leq 3$, $k \leq 4$ already records the full-horizon value.*

Survivability over a longer window is the conjunction of survivability over the shorter, giving the decreasing lattice; a monotone sequence on a finite lattice freezes; the instance’s length-independence and the window’s end give the two freezing statements, verified to horizon 40 by the verification script.

On the two-floor chance instance — states $\{0, 1\}$, actions $u \in \{\frac{2}{5}, \frac{3}{5}\}$, state 0 surviving exactly the lower action and state 1 exactly the higher — the one-step witness set is $\Gamma_1^\Pi = \{(1, 0), (0, 1)\}$ and $V_k(b_0) = \frac{1}{2}$ for all audited $k \leq 5$: the chance deficit $\delta = \frac{1}{2}$ is attained, and the deficit equals the minimal mass (Figure 5).

The asymmetric audit prices the deficit beyond the symmetric case. At $(z_0, T_{\text{obs}}) = (\frac{3}{2}, 2)$ with prior masses $w = \frac{3}{10}$ and $\frac{7}{10}$, the four blind two-step sequences survive $\{\frac{3}{10}, \frac{3}{10}, \frac{7}{10}, \frac{7}{10}\}$ (Table 2): every sequence loses a branch, so $V_2(b_0) = \frac{7}{10}$ and the deficit $\frac{3}{10}$ equals the minimal mass — the degenerate bound of Proposition 12 attained off the symmetric prior (Figure 2).

6. Certificates as deficit bounds

The obstruction certificates bound the deficit from below; their attaining witnesses are the deficit’s dual objects — computed lower-bound sides of the exact value, not surrogates for it.

Proposition 12 (deficit lower bounds). *Write $\Delta_k(b) = 1 - V_k(b)$ for the deficit, and let $\mathcal{W}_k^{\text{bel}}$ denote the k -step robust kernel in belief space: the family of beliefs whose support admits a jointly surviving declared-class sequence. Each companion certificate is a computable lower bound on the deficit. (i) Chance-constrained common action: if the one-step violation probabilities satisfy $\sum_x b(x) p(x, a) \geq \delta$ for every $a \in A$, then $\Delta_k(b) \geq \delta$ for every $k \geq 1$. (ii) Degenerate minimal mass: at the deterministic limit, if $b \notin \mathcal{W}_k^{\text{bel}}$ (the support admits no jointly surviving declared sequence), then $\Delta_k(b) \geq \min_{x \in \text{supp } b} b(x)$ — the two-floor instance attains $\delta = \frac{1}{2}$, and on the delayed grid at $(z_0, T_{\text{obs}}) = (\frac{3}{2}, 2)$ every blind sequence loses a branch,*

Table 1: The complete audited value table on the delayed grid: the exact $V_k(b_0)$ (values 1 or $\frac{1}{2}$) for all 16 grid states, three blind-window lengths, four horizons — 192 cell-horizon combinations, each regenerated by script and each equal to both the declared-class segment enumeration and the closed form of Proposition 5.

z_0	$T_{\text{obs}} = 1$				$T_{\text{obs}} = 2$				$T_{\text{obs}} = 3$			
	$k=1$	$k=2$	$k=3$	$k=4$	$k=1$	$k=2$	$k=3$	$k=4$	$k=1$	$k=2$	$k=3$	$k=4$
1.0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.2	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.3	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.4	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.5	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.6	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.7	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.8	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.9	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
2.0	1	1	1	1	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
2.1	1	1	1	1	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
2.2	1	1	1	1	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
2.3	1	1	1	1	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
2.4	1	1	1	1	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
2.5	1	1	1	1	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$

Table 2: The asymmetric audit at $(z_0, T_{\text{obs}}) = (\frac{3}{2}, 2)$, $b_0 = (\frac{3}{10}, \frac{7}{10})$: survived mass of each blind two-step sequence. Every sequence loses a branch at step 1 — at $z_0 = \frac{3}{2}$ the first move sends one branch to $\frac{1}{2} < 1$ —; the value is $\frac{7}{10}$ and the deficit $\frac{3}{10}$ equals the minimal mass.

Sequence	Survived mass	Lost branch
(+1, +1)	$\frac{3}{10}$	heavier branch, step 1
(+1, -1)	$\frac{3}{10}$	heavier branch, step 1
(-1, +1)	$\frac{7}{10}$	lighter branch, step 1
(-1, -1)	$\frac{7}{10}$	lighter branch, step 1

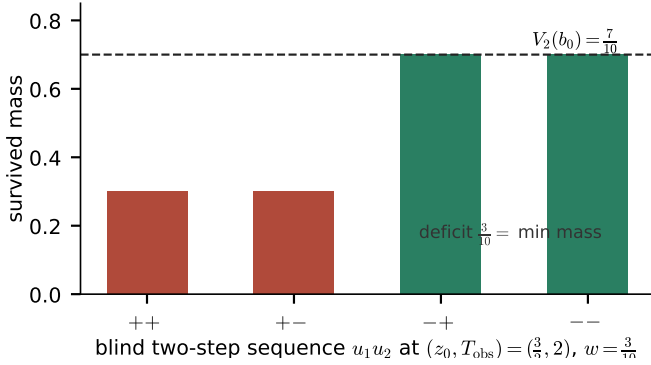


Figure 2: The asymmetric audit: survived mass by blind two-step sequence at $(\frac{3}{2}, 2)$ with $w = \frac{3}{10}$. The best sequences save the heavy branch only; the deficit $\frac{3}{10}$ is the minimal mass, attained.

so the deficit is at least the minimal mass $\frac{3}{10}$ of the audited asymmetric prior, attained (Table 2). (iii) Monotone deficit: $V_{k+1}(b) \leq V_k(b)$, so the deficit is non-decreasing in the horizon; on the certified nonviability region of the delayed grid the declared-class deficit is constant in k once $k > T_{\text{obs}}$ (and at every $k \geq 1$ on the common-action cells), while on the timing cells it steps from 0 at $k = 1$ to $\frac{1}{2}$ at $k \geq 2$ (Theorem 13d). For finite horizons the exact deficit $1 - \max_{\alpha \in \Gamma_k} \alpha^\top b$ is computable, and the bounds are witnesses of it, not surrogates.

7. Agreement on the delayed-regime class

The delayed hidden-regime instance of the calculus paper — regimes $\theta \in \{-1, +1\}$, blind window T_{obs} , declared hold class $\{u \equiv \pm 1\}$, prior $(\frac{1}{2}, \frac{1}{2})$ from z_0 — is the agreement test bed. The class declaration is part of the model: the blind-window policy may be required to hold one action throughout the window (the declared hold class), or may act sequentially without observation, or may act with open-loop time-variation.

Theorem 13 (certificate-value agreement). *On the audited grid ($z_0 \in \{1.0, \dots, 2.5\}$, $T_{\text{obs}} \in \{1, 2, 3\}$; 48 cells), over the declared hold class, and horizons $k \leq 4$: (a) the segment-form enumeration of the declared hold class (Proposition 7, segment form) returns exactly the closed form of Proposition 5 (on the audited grid the second indicator of the $k \leq T_{\text{obs}}$ line is vacuous, $z_0 + k \geq 2$; it is retained for branch symmetry). The class constraint is load-bearing here: a stagewise recursion over $A(\Pi) = \{\pm 1\}$ would compute the unrestricted value, which is 1 on the timing cells — the agreement is a statement about the segment enumeration; (b) the certificate partition coincides with the value boundaries: “viable” iff $V_k \equiv 1$; common-action-certified iff $V_1(b_0) < 1$; timing-certified iff $V_1(b_0) = 1$ but $V_{T_{\text{obs}}+1}(b_0) < 1$; (c) at the one-step horizon both mirror witnesses (1, 0) and (0, 1) attain the value $\frac{1}{2}$ below $z_0 = 2$, and the merged (1, 1) attains 1 at and above it; at horizon $k \leq T_{\text{obs}}$ the family changes at $z_0 = 1 + k$ — an off-grid consequence of the closed form (in particular $z_0 = 3$, not 2, for $k = 2$ with $T_{\text{obs}} \geq 2$), verified by the verification script beyond the audited grid, which stops at $z_0 = 2.5$; and read as a function of the continuous parameter z_0 the hold-class value has jump discontinuities exactly at $z_0 = 1 + \min(k, T_{\text{obs}})$ — a jump of exactly $\frac{1}{2}$, from $\frac{1}{2}$ to 1 — these loci being the certificate edges on the grid (kinks in the strict sense belong to the piecewise-linear value as a function of the belief, not of z_0); (d) on the nonviable cells the hold-class deficit equals $\frac{1}{2}$ at every horizon $k > T_{\text{obs}}$ and at every $k \leq T_{\text{obs}}$ with $k > z_0 - 1$, while it is 0 at horizons $k \leq z_0 - 1$ (on the audited grid: on the timing cells the deficit is 0 at $k = 1$ and $\frac{1}{2}$ at $k \geq 2$; on the common-action cells it is $\frac{1}{2}$ at every $k \geq 1$) — the minimal-mass bound attained with the minimal prior from the first horizon at which the falling branch exits.*

Derivation of the jump loci. Fix T_{obs} and k . On the closed form: if $k \leq T_{\text{obs}}$ the value is $\frac{1}{2}(\mathcal{K}[z_0 \geq 1 + k] + 1)$, the second indicator being identically one on the grid; it jumps by exactly

$\frac{1}{2}$ at $z_0 = 1 + k$. If $k > T_{\text{obs}}$ the value is $\frac{1}{2}(1 + \mathbb{I}[z_0 \geq 1 + T_{\text{obs}}])$, jumping by exactly $\frac{1}{2}$ at $z_0 = 1 + T_{\text{obs}}$. Both loci are $z_0 = 1 + \min(k, T_{\text{obs}})$; between jumps both expressions are constant in z_0 . The witness family changes exactly at the same loci: the attaining one-step vectors are both mirrors $(1, 0)$ and $(0, 1)$ below $z_0 = 2$ and the merged $(1, 1)$ at and above, and at horizon k the change sits at $z_0 = 1 + \min(k, T_{\text{obs}})$. $\square$

Theorem 14 (class lattice and realized gaps). *For the value of Definition 1, $\Pi \subseteq \Pi'$ implies $V_k^\Pi(b) \leq V_k^{\Pi'}(b)$. On the delayed instance with $\ell \geq 1$ and $z_0 \geq 1$:*

$$V^{\Pi_{\text{hold}}} \leq V^{\Pi_{\text{ol}}} = V^{\Pi_{\text{seq, blind}}} \leq V^{\Pi_{\text{obs}}},$$

the middle equality holding because no observation arrives inside the window, so sequential blind policies reduce to open-loop sequences; and $V^{\Pi_{\text{obs}}} = \mathbb{I}[z_0 \geq 1]$ (the controller knows θ at $t = 0$ and plays the rising control). The gaps are exact: $V^{\Pi_{\text{ol}}} - V^{\Pi_{\text{hold}}} = 1 - \max(w, 1 - w)$ exactly on $\{2 \leq z_0 < 1 + \ell\}$ — the value of flexibility — and $V^{\Pi_{\text{obs}}} - V^{\Pi_{\text{ol}}} = 1 - \max(w, 1 - w)$ exactly on $\{1 \leq z_0 < 2\}$ — the value of observation; at the symmetric prior each gap is $\frac{1}{2}$ on its cells and 0 elsewhere, and the two cell families are disjoint.

Inclusion monotonicity is immediate from the maximizing definition; the instance values are Theorem 6’s and the observed value is the rising computation; the gap arithmetic is $1 - \max(w, 1 - w) = \min(w, 1 - w)$.

Theorem 15 (class declaration). *The declared hold class and the unrestricted sequential blind class coincide at the one-step horizon. For horizons $k \geq 2$ they differ exactly where $2 \leq z_0 < 1 + \min(k, T_{\text{obs}})$ — on the audited grid ($z_0 \leq 2.5$) exactly the timing cells, since $1 + \min(k, T_{\text{obs}}) \geq 3$ there — and there open-loop time-variation within the blind window lifts the value to one: an alternating hold sequence keeps both branches at or above the floor whenever $z_0 \geq 2$, which no constant hold does (off the grid the hold class already attains one at $z_0 \geq 1 + k$: e.g. at $(z_0, T_{\text{obs}}, k) = (3.5, 3, 2)$ both classes attain one, so no difference). On the common-action cells $z_0 < 2$ the restriction costs nothing — no blind sequence saves both branches — and on the viable cells both classes attain one. On the audited grid the difference comprises exactly the 12 timing cells at each of $k = 2, 3, 4$: 36 cell-horizon combinations in all (Figure 3).*

Derivation of Theorem 15. At $k = 1$ both classes choose one blind action, so they coincide. Let $k \geq 2$ and write the two branch positions after a blind sequence $u_{1:\ell}$, $\ell = \min(k, T_{\text{obs}})$, as $z_0 \pm \sum_t u_t$. If $z_0 < 2$, some branch sits below $2 - (z_0 - 1)$ steps from its floor, and since the partial sums of any ± 1 -sequence move one branch down whenever they move the other up, one branch reaches its floor: no blind sequence saves both, both classes give $\frac{1}{2}$, no difference. If $z_0 \geq 1 + \min(k, T_{\text{obs}})$ the constant hold already survives the window on both branches (the falling branch exits only after $k > z_0 - 1$ steps) and both classes attain one. On the difference cells $2 \leq z_0 < 1 + \min(k, T_{\text{obs}})$: a constant hold drives the falling branch to $z_0 - \ell < 1$ (deficit), while the alternating sequence has partial sums in $\{0, \pm 1\}$, so both branches stay at or above $z_0 - 1 \geq 1$ through the window, and revelation at T_{obs} then regenerates both: the unrestricted class attains one while the declared class does not. This locates the difference setwise. $\square$

Remark 16 (the timing obstruction is class-scoped). The calculus paper’s timing certificates — $\sigma^*(B_0) = z_0 - 1$, with the certificate firing exactly when $T_{\text{obs}} > \sigma^*$ (equivalently $z_0 < 1 + T_{\text{obs}}$; viability at the deadline holds when $T_{\text{obs}} \leq \sigma^*$) — are statements about the declared hold-until-review class — the institutional reading of a sampled-review regulator. Theorem 15 quantifies the contrast: within the declared class the timing cells are nonviable with deficit $\frac{1}{2}$ at every horizon $k \geq 2$

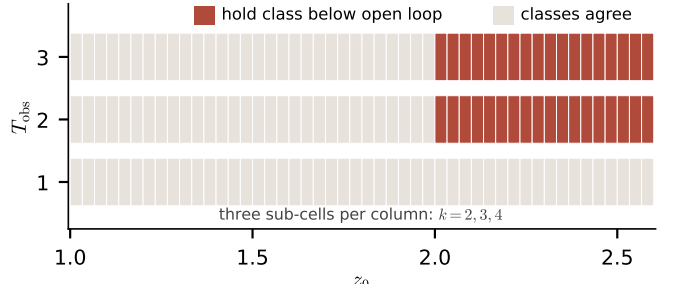


Figure 3: The class declaration on the audited grid: red cells mark cell-horizon combinations where the declared hold class and the unrestricted open-loop class differ — exactly the timing cells ($2 \leq z_0 < 1 + T_{\text{obs}}$) at horizons $k = 2, 3, 4$ (three sub-cells per column); 36 combinations in all. The classes agree at $k = 1$, on the common-action cells, and on the viable cells.

(and 0 at $k = 1$); under the unrestricted sequential blind class the same cells attain value 1 by oscillation, and on the audited grid no other cell differs. The timing obstruction therefore certifies exactly what it claims — nonviability *within the declared class* — and no statement of the companion papers is altered by this reading. The word “adaptive” is avoided for blind windows: while the window is blind, no observation updates the belief, and the only distinction available is hold-constant versus time-varying open-loop control.

8. Adaptive observation and hidden parameters

Observation is a design variable. Structural uncertainty — a hidden parameter resolved too late, rather than noisy data — is the obstruction: the state augments with a hidden parameter $\theta \in \Theta$, the belief lives on the product (x, θ) , and the obstruction is the emptiness of the joint floor-servable set over (x, θ) cells. On the declining instance the parameter $\theta \in \{-1, +1\}$ gates the correctable component of a $-\frac{1}{10}$ drift; the list of affected sustainability parameters — regeneration rates, climate sensitivity, institutional effectiveness — motivates the model class, and the paper verifies on exact instances.

Proposition 17 (probes and learning deadlines). *On the audited grid, with quantized control $u \in \{-1, 0, 1\}$ and hold-then-probe-then-act schedules: with θ observed, every cell is viable (16 of 16). Under one-probe endogenous learning with free probe timing (probe at the start), the probe reveals θ exactly — the branch separation is 2, so one observation splits the belief with no statistics — and survival forces $z_0 \geq 21/10$, independent of any deadline. Under the audited pinned-probe family — hold $u = 0$ to the probe horizon and probe at T_{learn} , so no earlier probe is available within the family — viability holds exactly when*

$$z_0 \geq \frac{21}{10} + \frac{T_{\text{learn}}}{10}, \quad T_{\text{learn}} = 0, 1, 2, 3, 4,$$

the pinned-probe learning-deadline identity: the deadline prices the forced late probe, and the kernel shrinks exactly linearly and antitone in it, $|K| = 5, 4, 3, 2, 1$ on the grid for $T_{\text{learn}} = 0, 1, 2, 3, 4$ (Table 3 and Figure 4). Two exact benchmarks bound the family from both sides: with free probe timing the threshold is flat at $21/10$, and with probe-free exogenous revelation at T_{learn} (hold, then the learned act; no probe) viability holds exactly when $z_0 \geq 1 + T_{\text{learn}}/10$ — kernel sizes 16, 15, 14, 13, 12 on the grid for $T_{\text{learn}} = 0, 1, 2, 3, 4$; the audited family sits between the two. A two-patch instance with unknown regeneration rate $g \in \{1, 2\}$ shows the same phenomenon on the audited

Table 3: The pinned-probe learning-deadline audit: the exact threshold $\frac{21}{10} + \frac{T_{\text{learn}}}{10}$ and the resulting kernel on the grid at each deadline.

T_{learn}	Threshold	$ K $	Kernel K on the grid
0	21/10	5	{2.1, 2.2, 2.3, 2.4, 2.5}
1	22/10	4	{2.2, 2.3, 2.4, 2.5}
2	23/10	3	{2.3, 2.4, 2.5}
3	24/10	2	{2.4, 2.5}
4	25/10	1	{2.5}

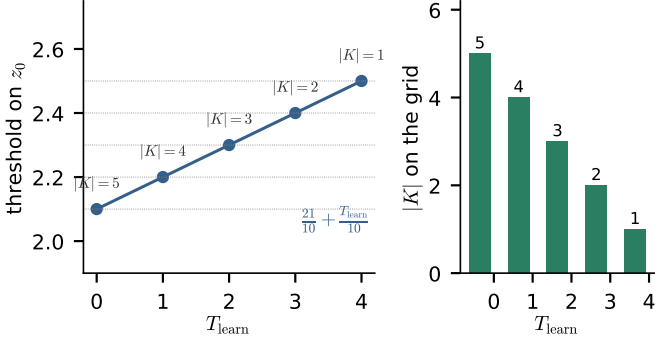


Figure 4: The pinned-probe learning-deadline identity: the viability threshold increases exactly linearly in T_{learn} (left), and the kernel on the grid shrinks linearly and antitone, $|K| = 5, 4, 3, 2, 1$ (right); the probe-free benchmark $1 + T_{\text{learn}}/10$ (kernels 16 down to 12) and the flat free-probe threshold $21/10$ bound it from below and above.

two-patch belief pair: unknown-regeneration nonviable, (x, g) -observed viable.

Proof. The schedule pins the probe to T_{learn} : within the family no earlier probe is admissible, so the free-probe and probe-free thresholds are not attainable in it. Before T_{learn} the drift is $-\frac{1}{10}$ (hold $u = 0$), so survival to the probe forces $z_0 - \frac{T_{\text{learn}}}{10} \geq$ the probe threshold; the probe $u = +1$ moves z by $-\frac{1}{10} + \theta$, which must stay safe on both branches, giving the threshold $\frac{21}{10}$ at $T_{\text{learn}} = 0$; after the probe the learned law $u = \theta$ has drift $-\frac{1}{10} + \theta^2 = \frac{9}{10} > 0$, so survival is indefinite. The branch positions $z_0 - \frac{1}{10}T_{\text{learn}} - \frac{1}{10} \pm 1$ differ by 2, so the post-probe reading determines θ uniquely. Antitonicity is the threshold identity; strictness on the grid is the computed counts 5, 4, 3, 2, 1. $\square$

9. Genuinely stochastic instances

Everything to this point has deterministic kernels and observations; the values are prior masses of surviving branches. This section opens the stochastic layer, where values cease to be two-valued and the min-mass bound ceases to be universal.

Proposition 18 (weighted-path deficit identity). *At the one-step horizon, with finite X , deterministic per-reading actions $a(y')$, and arbitrary normalized observations, the deficit is exactly*

$$\Delta_1(b) = \sum_x b(x) p_x, \quad p_x = \mathbb{P}(a(y') \text{ unsafe at } x | x),$$

the prior mass weighted by fatal-reading probabilities. The min-mass bound is the case $p_x \equiv 1$ (deterministic readings, no common action) and is not universal: in the two-state instance with a_i safe only at x_i and sensor error ε ($\mathbb{P}(\text{reads } j \neq i | x_i) = \varepsilon$), the deficit is ε for every prior; at $b = (\frac{1}{2}, \frac{1}{2})$, $\varepsilon = \frac{1}{20}$, the deficit $\frac{1}{20}$ is strictly below the minimal mass $\frac{1}{2}$.

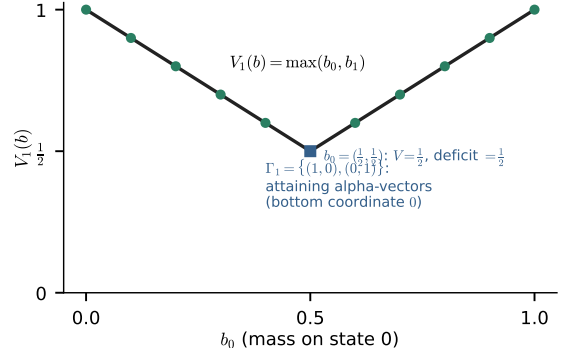


Figure 5: The two-floor chance instance: the one-step value is $\max(b_0, b_1)$, attained by the alpha-vectors of $\Gamma_1^\Pi = \{(1,0), (0,1)\}$ with zero bottom coordinate (green probes); at the symmetric prior the value is $\frac{1}{2}$ and the deficit equals the minimal mass.

Proposition 19 (noisy probes and the repetition trade-off). *On the declining instance (blind drift $-\frac{1}{10}$), pin a single probe to horizon t_p whose reading is correct with probability $1 - \varepsilon$ and flipped with probability ε , then act on the reading to horizon $k \geq t_p$. With $z_p = z_0 - t_p/10$:*

$$V_k(b_0) = (1 - \varepsilon) \mathbb{I}[z_p \geq 1] + \varepsilon \mathbb{I}[z_p \geq 1 + \frac{11}{10}(k - t_p)],$$

exactly (the correct reading acts under drift $\frac{9}{10}$, the flipped reading under drift $-\frac{11}{10}$). The values realize exactly three levels $\{0, 1 - \varepsilon, 1\}$ — genuinely non-two-valued — and the min-mass bound survives here as a bound (the deficit ε or 1 dominates the post-probe minimal mass ε) while failing as an identity. With three pinned probes and majority reading, the wrong-majority probability is $3\varepsilon^2(1 - \varepsilon) + \varepsilon^3 = \varepsilon^2(3 - 2\varepsilon) - \frac{7}{250}$ at $\varepsilon = \frac{1}{10}$; against $\frac{1}{10}$ for one probe — and the same display holds with $p = \frac{7}{250}$ in place of ε : each additional probe step costs $\frac{1}{10}$ of floor margin to the blind drift, the exact explore-exploit exchange at this scale.

Proof. Condition on $(\theta, \text{reading})$: the four joint cases carry probabilities $\frac{1}{2}(1 - \varepsilon)$ and $\frac{1}{2}\varepsilon$ per branch; the correct reading acts under drift $-\frac{1}{10} + \theta^2 = \frac{9}{10}$ and survives from $z_p \geq 1$; the flipped reading acts under drift $-\frac{1}{10} - 1 = -\frac{11}{10}$ and survives exactly when $z_p - \frac{11}{10}(k - t_p) \geq 1$; summing gives the display. For the majority family, the wrong-majority event is two or three flipped readings, of probability $3\varepsilon^2(1 - \varepsilon) + \varepsilon^3$; the falling continuation is as above. The identity's proof is the law of total probability over readings at the one-step horizon. $\square$

10. Distributionally robust interpolation

The expected and adversarial readings of Remark 9 are the two ends of one exact family. Fix the contamination ambiguity set $\mathcal{D}_\rho = \{q = (1 - \rho)p + \rho r : r \in \Delta(Y)\}$ at radius $\rho \in [0, 1]$.

Proposition 20 (contamination interpolation). *Let $\mathcal{D}_\rho$ be as above and define $V_{k+1}^\rho(b) = \max_a \inf_{q \in \mathcal{D}_\rho} \sum_y q(y) V_k^\rho(\tau(b, a, y))$. Then: (i) the inner infimum is exact,*

$$\inf_{q \in \mathcal{D}_\rho} \sum_y q(y) v_y = (1 - \rho) \sum_y p(y) v_y + \rho \min_y v_y,$$

the adversary spending the whole contamination ρ on the worst outcome; (ii) $V_k^\rho(b) = (1 - \rho) V_k^0(b) + \rho V_k^1(b)$ at every stage: the interpolation is pointwise exact, with $\rho = 0$ the expected value of Definition 1 and $\rho = 1$ the adversarial reading; (iii) V_k^ρ is concave and piecewise linear in b at every ρ , so the exact-arithmetic discipline of Proposition 7 carries over verbatim.

The infimum of a linear functional over the contamination set is attained at a vertex $r = \delta_{y^*}$, giving (i); (ii) follows by induction with (i) at every stage, since the adversary’s optimal contamination is stage-wise; concavity is preserved under pointwise mixtures of concave piecewise-linear functions. On the delayed class with prior contamination $q(+1) = (1 - \rho)w + \rho r$, the hold-class value is exactly

$$V_k^{\rho, \Pi^{\text{hold}}}(b_0) = \begin{cases} 0, & z_0 < 1, \\ 1, & z_0 \geq 1 + \ell, \\ (1 - \rho) \max(w, 1 - w), & 1 \leq z_0 < 1 + \ell, \end{cases}$$

the adversary pouring the contamination onto a falling branch; at the symmetric prior the three ρ -levels are 0, $\frac{1-\rho}{2}$, and 1 (verified by exact enumeration on the swept grid). This proposition is the formal home of worst-case wording: an ambiguity-minded reviewer may read every expected value here at any ρ , and every table carries the $\rho = 0$ column of a one-parameter family.

Proposition 21 (probe-count bound). *Let $n = 2m + 1$ pinned probes with per-probe error $\varepsilon < 1/2$ be read by majority. The wrong-majority probability satisfies*

$$p_{\text{wrong}}(n) \leq (4\varepsilon(1 - \varepsilon))^{n/2} \leq e^{-2n \text{sep}^2}, \quad \text{sep} = \frac{1}{2} - \varepsilon,$$

so $n \geq \ln(1/\delta)/(2\text{sep}^2)$ probes achieve wrong probability $\leq \delta$: the probe count grows like $\log(1/\delta)$ over the squared separation, and on the declining instance each probe step costs $\frac{1}{10}$ of floor margin to the blind drift (the exploration-safety exchange of Proposition 19, now with its rate). At $\varepsilon = \frac{1}{10}$ the bound reads $p_{\text{wrong}}(n) \leq (3/5)^n$, verified exactly at $n = 1, 3, 5, 7, 9$ (the exact values $\frac{1}{10}, \frac{7}{250}, \frac{107}{12500}, \frac{341}{125000}, \frac{22273}{25000000}$).

The bound is the standard misselection probability of the binomial head: $p_{\text{wrong}}(n) \leq (4\varepsilon(1 - \varepsilon))^{n/2}$ by pairing each wrong-majority string with its complement and Stirling’s estimate, and $4\varepsilon(1 - \varepsilon) = 1 - 4\text{sep}^2 \leq e^{-4\text{sep}^2}$.

Proposition 22 (general additive deadline law). *Let the hidden parameter enter the drift additively: $\dot{z} = d(u, \theta) = c(u) + h(u)\theta$ with $\theta \in \Theta$ finite, floor $z \geq 1$, and consider the hold-probe-learn schedule family: hold u_0 to the probe horizon T (the probe pinned there, no earlier probe in the family), probe with u_p , then play $u^*(\theta)$ with $\min_{\theta} d(u^*(\theta), \theta) = \mu > 0$. Set $d_0 = \max_{\theta} (-d(u_0, \theta))$ and $e_p = \max_{\theta} (-d(u_p, \theta))$. Then indefinite viability holds exactly when*

$$z_0 \geq 1 + T d_0 + e_p,$$

the floor plus drift times deadline plus probe excursion; after the learned act the positive drift μ preserves safety forever, so the deadline identity is the whole answer. The audited instance has $d_0 = \frac{1}{10}$, $e_p = \frac{11}{10}$, giving $\frac{21}{10} + \frac{T}{10}$; a second instance with drift $-\frac{3}{20}$ and excursion $\frac{21}{20}$ gives $\frac{41}{20} + \frac{3T}{20}$ — both verified exactly at all swept (z_0, T) .

Necessity: some θ realizes the hold displacement d_0 and the probe excursion e_p at the adversarial stages (the maxima are attained), dropping the state to $z_0 - T d_0 - e_p$, below the floor exactly when the display fails. Sufficiency: every θ keeps the state at or above the floor through hold and probe, and the learned drift $\mu > 0$ then lifts it away from the floor forever.

11. Verification methods

Every value, bound, identity, threshold, figure, and tabulated entry in this paper is regenerated by deterministic scripts in exact integer and rational arithmetic, standard library only. Three per-system scripts cover the audited instances, and one master entry point chains them. The belief-state campaign

script runs sixteen check families on the stochastic values (the two-floor instance, piecewise-linearity, the deterministic limit, the 48-cell agreement, the partition coincidence, the witness family, the deficit identity, the unrestricted class, rationality of all 384 campaign alpha-vectors, deficit monotonicity, the oscillation identity, model completeness, the selector split, dimension consistency, the jump loci, and the declarations), chaining two per-instance scripts of twenty-one further checks. The stochastic selector script re-derives every selector claim in fifteen check families (S1–S15: the two-floor backup, eleven-belief piecewise linearity, degeneration against brute force, 48-cell closed-form agreement, partition coincidence, witness-family loci, the exact deficit on all 42 nonviable cells, the class declaration by brute force, rationality of the 384 vectors, monotonicity, the oscillation identity to horizon 24, model completeness, the selector split, dimension padding, and the one-sided jump values). The hidden-parameter script runs six check families on the adaptive-observation instance (theta observed on all 16 cells, the probe threshold $\frac{21}{10}$ as a branchwise identity, the deadline identity at $T_{\text{learn}} = 0, \dots, 4$, injective revelation, the antitone kernel chain, and the unknown-regeneration two-patch instance). The master script for this paper chains the three, then recomputes independently from the definitions: the closed-form values on the whole 16×12 grid by direct branch-mass enumeration, the resulting census (36 unit values and 156 half values on the declared class), the identity of the class-difference set with the region $2 \leq z_0 < 1 + \min(k, T_{\text{obs}})$, checked on an extended grid to $z_0 = 4.0$ and equal to the timing-cell combinations on the audited grid at $k \geq 2$, by brute-force enumeration over blind sequences; the 384-vector type census (156, 156, 72) and its deduplicated 348-vector form; the hold-class deficit profile on all 48 cells (0 before the falling-branch exit, $\frac{1}{2}$ after); the lost-branch step of the asymmetric audit (step 1 for all four sequences, heavier branch under $u_1 = +1$, lighter under $u_1 = -1$); the pinned-probe learning-deadline chain $\frac{21}{10} + \frac{T_{\text{learn}}}{10}$ with the free-probe (21/10, flat) and probe-free ($1 + T_{\text{learn}}/10$, kernels 16 down to 12) benchmarks; and every headline number asserted present in the text and bound to the chained scripts’ certified families. Two independent paths back the closed forms: a brute-force path that enumerates trajectories straight from the definitions (all holds and all open-loop sequences, pathwise survival) over a structured sweep of 2268 triples (z_0, w, ℓ) including $z_0 < 1$, and a seeded randomized battery of 500 further cases; the survivable-set antichain, the class ordering and its gaps, the stabilization freeze (spot enumeration to horizon 15 and value agreement to horizon 40), the weighted-path identity, the sensor counterexample, and the noisy-probe and majority closed forms are each re-derived by independent enumeration; the contamination lemma and the delayed-class ρ -closed form are checked against vertex enumeration, the probe-count bound at $n = 1, 3, 5, 7, 9$ in exact rationals, and the additive deadline law on both instances.

12. Delimitations and outlook

The semantics here is stochastic expectation: the value is a survival *probability*, and the robust quantifiers of the companion papers are recovered only at the deterministic degeneration, which is a change of operator (Remark 9); the stochastic and robust readings of a genuinely stochastic plant differ, and this paper does not conflate them. The timing layer’s statements, here and in the companion papers, are class-scoped (Remark 16). No learning theory is claimed beyond the one-probe exact split: Bayesian updating, repeated experiments, and exploration-exploitation trade-offs are out of scope; the deadline identity is for the pinned-probe schedule family (hold to the probe horizon, probe, learned act), with the free-probe and probe-free benchmarks stated alongside, and other schedules are not classified; the parameter θ enters additively, and state-

dependent parameter dynamics belong to the hybrid companion’s setting. The alpha-set growth is exponential in the worst case — and finite-horizon partially observable problems are PSPACE-complete, their no-observation variant NP-complete (Papadimitriou and Tsitsiklis, 1987) — so the audited two-witness families are structural, not representative. The contamination interpolation of Section 10. is the formal home of worst-case wording; the moment- and distance-based ambiguity families (piecewise-linear values persist there; Nakao, Jiang, and Shen, 2021) are the natural next ring and are not developed here. Scaling past the audited instances — exact point-based value iteration with rational solvers, survivable-set compression over the antichain of Proposition 4, and medium instances with several hidden parameters — the probe-count asymptotics of the repetition trade-off, the general learning-deadline law for additively entering hidden parameters, and a proof-assistant formalization of the closed forms are the natural next steps; the deficit-bound reading of Proposition 12 remains the interface through which this theory attaches to the certificate families already established, and the exact rational values of the audited campaign are the benchmark any relaxation must meet.

13. Conclusion

The belief-state safety value carries the obstruction calculus into the probabilistic setting without loss of exactness. The support identity ties the value-one level to the viable-set recursion; rational alpha-vectors keep every audited value exact, and the complete value table shows the two-valued structure the certificates predict; the certificates survive as deficit lower bounds, attained on natural instances on both the symmetric and the asymmetric prior; the audited delayed-regime class agrees with its closed form cell by cell, with the class declaration responsible for exactly the 36 timing combinations on the grid; and the adaptive-observation audit prices late resolution exactly — for the pinned-probe family a linear learning deadline on the viable set, $|K| = 5, 4, 3, 2, 1$, bracketed by the flat free-probe threshold and the probe-free benchmark — while the stochastic layer prices observation noise exactly: a weighted-path deficit identity, a sensor instance breaking the min-mass bound’s universality, and a repetition trade-off at 7/250 per extra probe pair. The observation layer, not the forecast layer, sets the value; the design rules are the same ones the deterministic audits impose, now with a price attached and every number in the tables open to script-level re-inspection.

References

- Abaee, A. An obstruction calculus for viability under incomplete observation (submitted).
- Abaee, A., 2026a. Robust viability of the 2J3KL limit reference point under a surplus-production map: policy scoring, expansion, and when catch cannot help. Zenodo. <https://doi.org/10.5281/zenodo.22552060>
- Bertsekas, D.P., Shreve, S.E., 1978. Stochastic Optimal Control: The Discrete-Time Case. Academic Press, New York.
- Åström, K.J., 1965. Optimal control of Markov processes with incomplete state information. J. Math. Anal. Appl. **10**, 174–205.
- Alshiekh, M., Bloem, R., Ehlers, R., Könighofer, B., Niekum, S., Topcu, U., 2018. Safe reinforcement learning via shielding. Proc. AAAI Conf. Artif. Intell. **32**(1).
- Chatterjee, K., Doyen, L., Henzinger, T.A., 2009. Qualitative analysis of partially-observable Markov decision processes. In: Int. Symp. Math. Found. Comput. Sci. (MFCS 2009). LNCS, vol. 5734, pp. 635–646.
- Nakao, H., Jiang, R., Shen, S., 2021. Distributionally robust partially observable Markov decision process with moment-based ambiguity. SIAM J. Optim. **31**(1), 461–488.
- Papadimitriou, C.H., Tsitsiklis, J.N., 1987. The complexity of Markov decision processes. Math. Oper. Res. **12**(3), 441–450.

Smallwood, R.D., Sondik, E.J., 1973. The optimal control of partially observable Markov processes over a finite horizon. Oper. Res. **21**(5), 1071–1088.

Declarations

Funding

None.

Competing interests

None.

Data availability

No external data were used.

Code availability

The verification script `paper2_probabilistic_sufficiency_v4_verification.py` — which chains the per-system scripts `paper2_belief_state_v2_verification.py`, `paper2_stochastic_selector_v2_verify.py`, and `hidden_parameter_learning_v1_verify.py` — and the figure-and-table generation script `paper2_belief_state_figures.py` are publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`, at the commit pinned to this article’s submission); they reproduce all values, bounds, thresholds, figures, and tabulated entries verbatim.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.